

Faculty of Computing

BSc (Hons) in Information Technology Year 4 Semester 1 & 2

SE4041 - Mobile Application Design and Development (MADD) **Assignment 02 – Swift**

Assignment - 2

SE4041 – Mobile Application Design and Development Semester 2, 2026

Marks – Total Marks allocated 25% Deadline – November 2026

Part A – 15% of the Marks

Design, develop, and deploy a fully functional iOS app that incorporates advanced UI components, leverages modern iOS frameworks, and integrates at least one emerging technology (e.g., machine learning, augmented reality, IoT integration). The app should have a minimum of **three navigable screens**, employ sophisticated data persistence strategies, and adhere to best practices in mobile application development.

Requirements:

1. Purpose & Target Audience:

- o Define a clear purpose for your app that solves a real-world problem or offers a unique service.
- o Identify and justify your target audience, ensuring the app's features and design meet their needs.

2. Advanced UI Elements:

- o Incorporate advanced user interface elements such as custom animations, transitions, gesture recognizers, and adaptive layouts.
- o Ensure the UI is responsive and provides an intuitive user experience across different devices and orientations.

3. SwiftUI Components & Customization:

- o Utilize a variety of SwiftUI components, demonstrating advanced customization and theming.
- o Implement at least one custom SwiftUI component or modifier.

4. Navigation & Architecture:

- o Implement a robust navigation structure using UINavigationController or similar patterns.

5. Data Persistence & Networking:

- o Implement data persistence using Core Data, Realm, or other advanced methods.

6. Integration of Emerging Technologies:

- o Incorporate at least one advanced iOS library or framework such as CoreML for machine learning, ARKit for augmented reality, HealthKit, or HomeKit.
- o Demonstrate meaningful integration that enhances the app's functionality.

Appropriate use of any one of these libraries

- (a) <https://developer.apple.com/machine-learning/>
- (b) <https://developer.apple.com/augmented-reality/>
- (c) <https://developer.apple.com/maps/>
- (d) <https://developer.apple.com/health-fitness/>
- (e) <https://developer.apple.com/games/>
- (f) <https://developer.apple.com/education/>
- (g) <https://developer.apple.com/localization/>

7. Visual Design & Branding:

- o Develop a cohesive visual design that includes custom icons, branding elements, and a consistent color scheme.
- o Ensure accessibility standards are met, including support for Dark Mode and Dynamic Type.

8. Testing, Debugging & Performance Optimization:

- o Conduct testing, including unit tests, UI tests, and user acceptance testing.
- o Optimize app performance, focusing on memory management, load times, and responsiveness.

9. Documentation & Reporting:

- o Provide comprehensive documentation, including a user guide, technical documentation, and an in-depth report.
- o The report should discuss design decisions, development challenges, testing results, and reflections on the development process.

10. Deployment & Presentation:

- o Present your app in a live demonstration, highlighting key features and innovations.

Part A: Advanced iOS App (15%) - Rubric

Criteria	Marks	Description
App Functionality	15	The app works flawlessly, with all features implemented as intended. Demonstrates advanced functionality and robustness.
Advanced UI & UX Design	15	Exceptional UI/UX design, including advanced UI elements, custom animations, and responsive layouts. Provides an outstanding user experience.
Code Quality & Architecture	15	High-quality, well-organized code following best practices.
Integration of Emerging Technologies	15	Successfully integrates advanced technologies (e.g., ML, AR). The integration enhances app functionality in meaningful ways.
Testing & Optimization	10	Comprehensive testing conducted. App is optimized for performance, with efficient memory management and fast load times.
Documentation & Reporting	10	Detailed and professional documentation. In-depth report covering all aspects of design, implementation, testing, and reflections on the development process.
Innovation & Creativity	10	Demonstrates exceptional creativity and innovation. The app offers unique features or approaches that set it apart.
Presentation & Deployment Readiness	10	Presentation is professional, clearly demonstrating the app's features and value.

Part B – Innovative Application for VisionOS or tvOS (10% of the Marks)

Develop a sophisticated prototype application for VisionOS or tvOS that leverages the unique capabilities of these platforms. The prototype should demonstrate innovative use of the platform's features and provide a compelling user experience. Think creatively and propose an application that could have significant impact or market potential.

Requirements:

1. Innovative Concept Development:

- o Propose a novel app concept tailored specifically for VisionOS or tvOS, addressing a unique need or offering a new entertainment experience.
- o Find and describe existing Apps which are similar

2. Advanced Prototype Development:

- o Develop a functional prototype that showcases core features, user interactions, and platform-specific capabilities.
- o Utilize advanced features such as spatial computing for VisionOS or multi-user experiences for tvOS.

3. User Experience & Interaction Design:

- o Design intuitive and engaging user interactions, leveraging the input methods and interaction models of the chosen platform.
- o Ensure the prototype adheres to platform design guidelines and best practices.

4. Technical Implementation:

- o Demonstrate technical proficiency by integrating complex features, optimizing performance, and ensuring stability.
- o Implement code using appropriate architectural patterns and coding standards.

5. Pitch Presentation:

- o Prepare a professional pitch presentation aimed at potential investors or stakeholders.
- o The presentation should include the app's value proposition, target market, revenue model, and go-to-market strategy.
- o Present your prototype during the viva, showcasing its features and explaining its potential impact.

Part B: VisionOS/tvOS App - Rubric (10% of Marks)

Criteria	Marks	Description
Concept Innovation	25	The app concept is highly innovative, with strong market potential. Addresses a unique need or provides a new kind of experience.
Prototype Functionality	25	The prototype demonstrates core functionalities effectively, showcasing advanced features of the platform.
User Experience Design	15	Provides an engaging and intuitive user experience, leveraging platform-specific interaction models.
Technical Proficiency	15	Demonstrates technical expertise in implementing platform-specific features and optimizing performance.
Pitch Presentation	20	Compelling pitch that effectively communicates the app's value proposition, market potential, and business strategy. Presentation is professional and persuasive.

Submission Guidelines:

- **Individual Assignment:** This assignment must be completed individually.
- **Version Control:** Use a GitHub repository for each project. Ensure the repositories are well-organized, with clear commit messages, branches for feature development, and properly managed pull requests if applicable.
- **Code Documentation:** All code should be well-commented, following industry standards. Include a README file with instructions on how to build and run your app.
- **AI-Assisted Development:** If you use AI tools like GitHub Copilot or ChatGPT, document the interactions, including prompts and responses, and explain how they assisted your development process.
- **Original Work:** Ensure all code, designs, and content are your original work. Properly cite any third-party resources or libraries used.
- **Demonstration & Viva:** Be prepared to present and defend your projects during a viva session. You will be asked to discuss your design choices, implementation details, and how you addressed challenges.
- If you are referring to tutorials on the web, we still expect you to write your own code. If the code that you submitted is similar to code available in the internet, we would suspect it to be plagiarized.

This assignment is an opportunity to demonstrate your mastery of mobile application development, your ability to innovate, and your readiness to enter the professional world. Focus on delivering high-quality, polished applications that you would be proud to showcase in a professional portfolio.

Good luck!

Part A: Advanced iOS App Rubric (Detailed) — 15%

Criteria	Excellent	Very Good	Good	Average	Needs Improvement
App Functionality (Max 15 Marks)	App works flawlessly with all required features fully implemented. Demonstrates advanced functionality, reliability, and robustness.	App works very well with most features fully implemented. Minor issues may exist but do not affect overall use.	App works with main features implemented. Some minor bugs or incomplete areas are present.	App has basic functionality, but several features are incomplete or unstable.	App has major functionality issues, missing features, or does not run properly.
Advanced UI & UX Design (Max 15 Marks)	Exceptional UI/UX with advanced UI elements, custom animations, smooth transitions, responsive layouts, and excellent usability.	Very good UI/UX with strong design, good responsiveness, and some advanced interface features.	Good UI/UX with clear navigation and acceptable visual design, but limited advanced features.	Basic UI/UX with some usability issues, inconsistent layouts, or limited responsiveness.	Poor UI/UX with confusing navigation, weak design, or lack of responsiveness.
Code Quality & Architecture (Max 15 Marks)	Code is well-structured, clean, efficient, reusable, and follows best practices and suitable architectural patterns.	Code is organized and mostly follows best practices, with minor improvements needed.	Code is understandable and functional, but structure, reuse, or architecture could be improved.	Code works but is poorly organized or inconsistent in style and structure.	Code is difficult to understand, poorly structured, incomplete, or does not follow good practices.
Integration of Emerging Technologies (Max 15 Marks)	Successfully integrates an advanced iOS framework such as CoreML, ARKit, HealthKit, Maps, or similar in a meaningful and innovative way.	Integrates an emerging technology well, with clear relevance to app functionality.	Uses an emerging technology, but the integration is basic or only partly meaningful.	Attempts to include an emerging technology, but implementation is limited or unclear.	No meaningful emerging technology integration is included.

Criteria	Excellent	Very Good	Good	Average	Needs Improvement
Testing & Optimization (Max 10 Marks)	Comprehensive testing is completed, including unit/UI/user testing. App is optimized for performance, memory use, speed, and responsiveness.	Good testing and optimization are evident, with only minor issues remaining.	Some testing has been conducted, and performance is acceptable.	Limited testing and optimization; noticeable performance or stability issues exist.	Little or no testing is evident; app has major performance or stability problems.
Documentation & Reporting (Max 10 Marks)	Documentation is detailed and professional, including user guide, technical documentation, design decisions, challenges, testing results, and reflections.	Documentation is clear and covers most required areas with good detail.	Documentation covers the main areas but lacks depth in some sections.	Documentation is basic, incomplete, or lacks important details.	Documentation is missing, unclear, or does not explain the project properly.
Innovation & Creativity (Max 10 Marks)	App demonstrates exceptional creativity, originality, and unique features that make it stand out.	App shows strong creativity and includes some unique or interesting features.	App shows some creativity, but the idea or features are fairly common.	Limited creativity; app is mostly basic or similar to existing examples.	Little or no creativity or innovation is demonstrated.
Presentation & Deployment Readiness (Max 10 Marks)	Presentation is professional, confident, and clearly demonstrates the app's features, value, and readiness for deployment.	Presentation is clear and well-organized, with good demonstration of app features.	Presentation explains the app adequately but lacks depth or polish.	Presentation is basic, unclear in parts, or does not fully demonstrate the app.	Presentation is poor, incomplete, or the app is not ready to demonstrate.

Part B: VisionOS / tvOS App Rubric — 10%

Criteria	Excellent	Very Good	Good	Average	Needs Improvement
Concept Innovation (Max 25 Marks)	App concept is highly innovative, original, and has strong market potential. It addresses a unique need or creates a new experience.	Concept is creative and relevant, with clear value and some market potential.	Concept is good but not highly original; it addresses a clear need or experience.	Concept is basic, with limited originality or unclear market potential.	Concept is weak, unclear, copied, or lacks innovation.
Prototype Functionality (Max 25 Marks)	Prototype effectively demonstrates core features and advanced VisionOS/tvOS platform capabilities.	Prototype works well and demonstrates most key features clearly.	Prototype demonstrates basic core features, but advanced platform use is limited.	Prototype is partially functional, with several missing or incomplete features.	Prototype is non-functional, very incomplete, or fails to demonstrate the concept.
User Experience Design (Max 15 Marks)	User experience is highly engaging, intuitive, and makes excellent use of platform-specific interaction models.	UX is clear, engaging, and mostly follows platform guidelines.	UX is acceptable and understandable, but lacks polish or deeper platform-specific design.	UX is basic and may have usability or interaction issues.	UX is confusing, poorly designed, or does not suit the chosen platform.
Technical Proficiency (Max 15 Marks)	Demonstrates strong technical skill, stable implementation, optimized performance, and appropriate coding standards.	Shows good technical ability with mostly stable and well-implemented features.	Shows adequate technical skill, though some implementation or performance issues exist.	Shows limited technical skill, with noticeable bugs or weak implementation.	Technical implementation is poor, unstable, incomplete, or not suitable for the platform.
Presentation (Max 20 Marks)	Pitch is professional, persuasive, and clearly explains value proposition, target market, revenue model, go-to-market strategy, and impact.	Pitch is clear and convincing, covering most business and product aspects well.	Pitch explains the concept and value adequately but lacks detail in some areas.	Pitch is basic and misses important business or market details.	Pitch is unclear, incomplete, unprofessional, or does not communicate the app's value.